

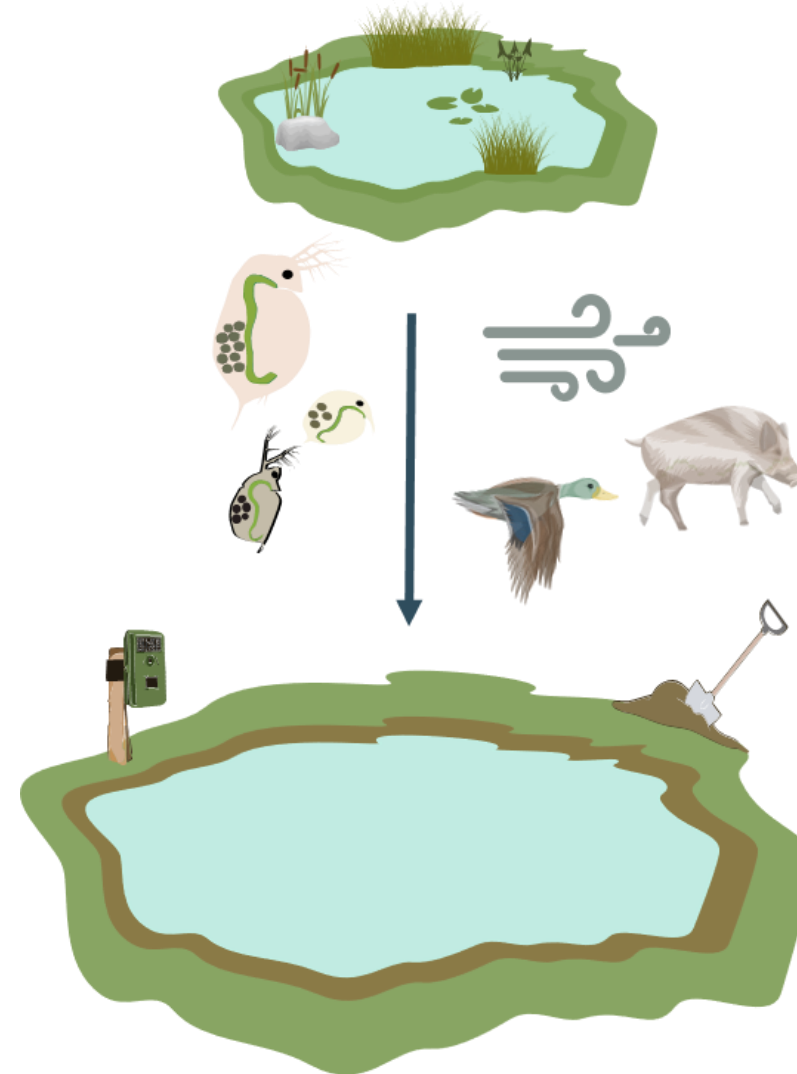
Evolving metacommunities

Bram Janssens



My research

- Look at colonisation patterns of water flea communities into newly created ponds
- Research how metacommunities develop and evolve through time
- Using complex cameratrap data in research



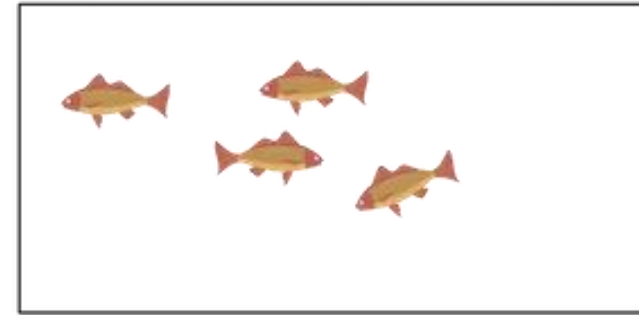
Learning objectives

- What is difficult about community studies?
- What is the concept of a metacommunity?
- How does evolution play a role in communities of multiple species?
- What is the interplay between ecological and evolutionary processes at the level of communities?

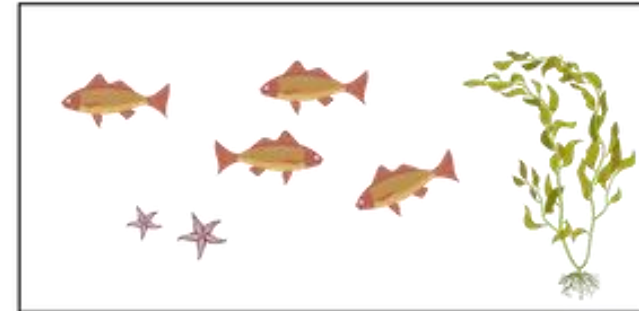
Introduction

What is a community?

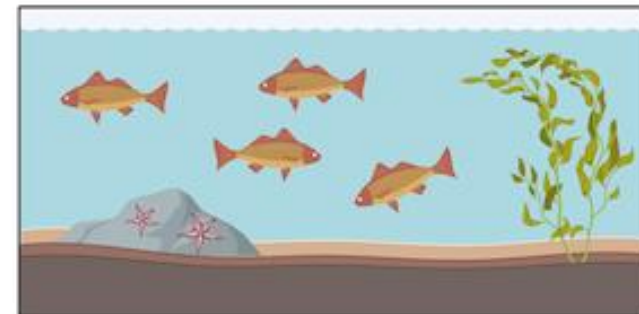
- **Population:** groups of individuals of the same species that live together and reproduce
- **Community:** groups of different species living together
- **Ecosystem:** all living organisms and non-living material in the same geographical space



population



community



ecosystem

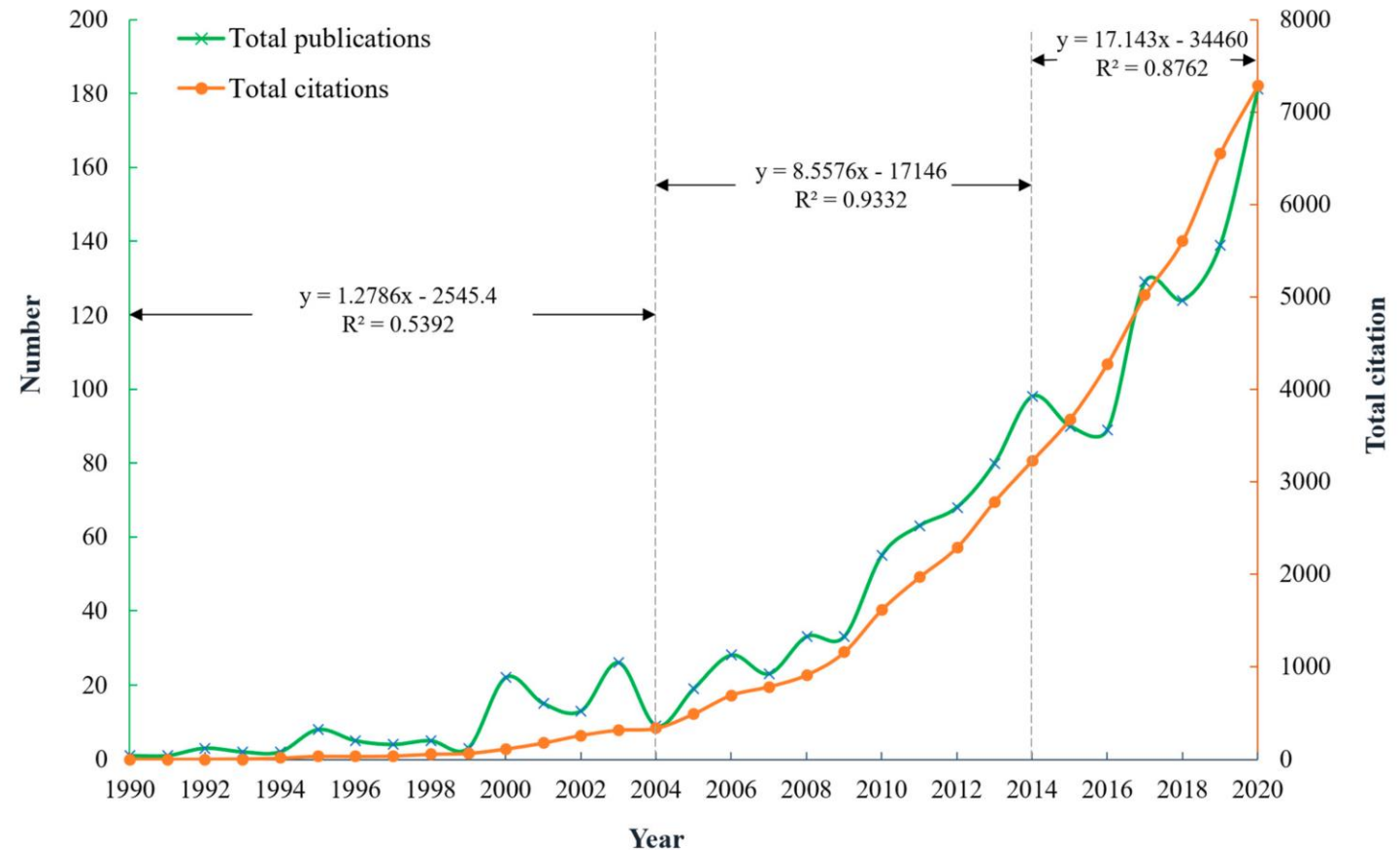
What is a community?



Community ecology research

- Strong increase in number of publications in last years
- Remains low compared to population ecology research

➤ Why?



Community ecology research



Advantages	Challenges

Community ecology research



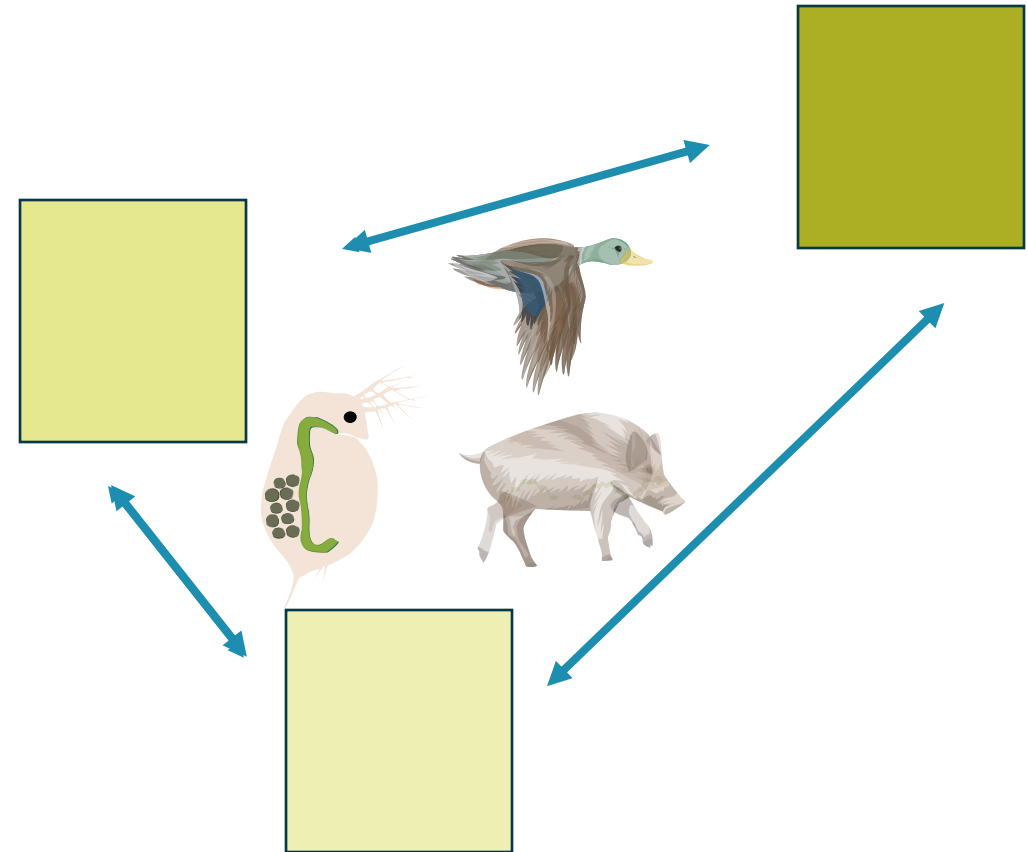
Advantages	Challenges
Broader perspective	Complexity and Interconnectedness
Biodiversity and Stability	Isolating Factors and Cause-and-Effect Relationships
Invasive Species and climate change	Defining Community Boundaries
Ecological Restoration and conservation	Data Collection and Analysis

Communities interact with each other...

➤ Metacommunities

*A set of local communities linked by the **dispersal** of multiple, potentially interacting species.*

Dispersal= the movement of individuals or propagules away from their birth or breeding site to a new location where they may establish and reproduce



At different scales for different species groups

Islands are terrestrial patches within an aquatic matrix

- **Avian dispersal**
- Bird dispersal
- Wind dispersal
- Rafting
- ...



At different scales for different species groups

Islands are terrestrial patches within an aquatic matrix

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At different scales for different species groups

Islands are terrestrial patches within an aquatic matrix

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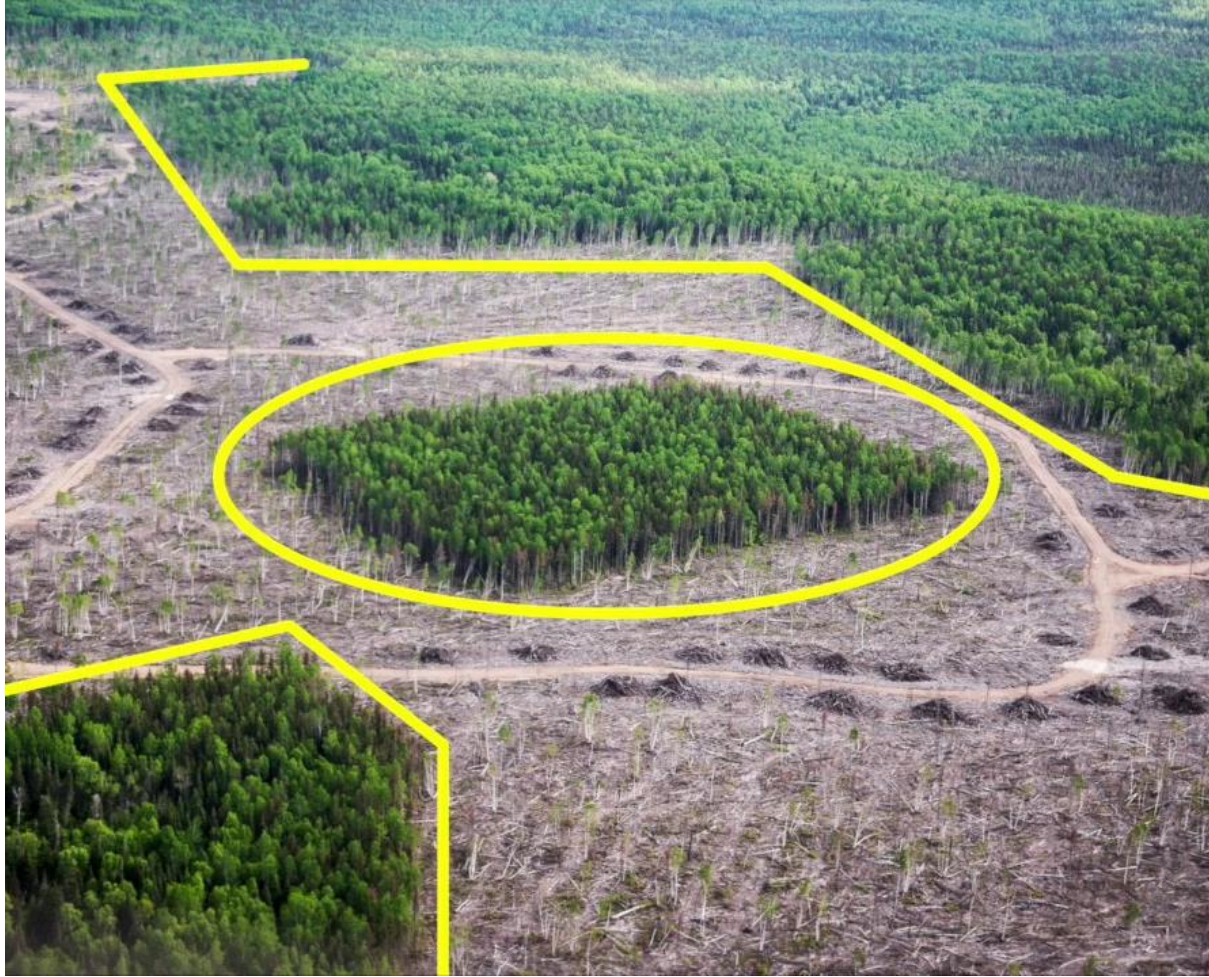
At different scales for different species groups

Islands are terrestrial patches within an aquatic matrix

- Avian dispersal
- Bird dispersal
- Wind dispersal
- **Rafting**
- ...



At different scales for different species groups



At different scales for different species groups



Metacommunity ecology research

Key questions:

- How do regional scale spatial dynamics affect the outcome of ecological interactions in local communities?
- How does the interplay between dispersal and local interactions determine metacommunity properties at the regional scale?

Metacommunity ecology research

- Mainly studied through theoretical models and concepts
 - Metacommunity concept by **Leibold 2004**

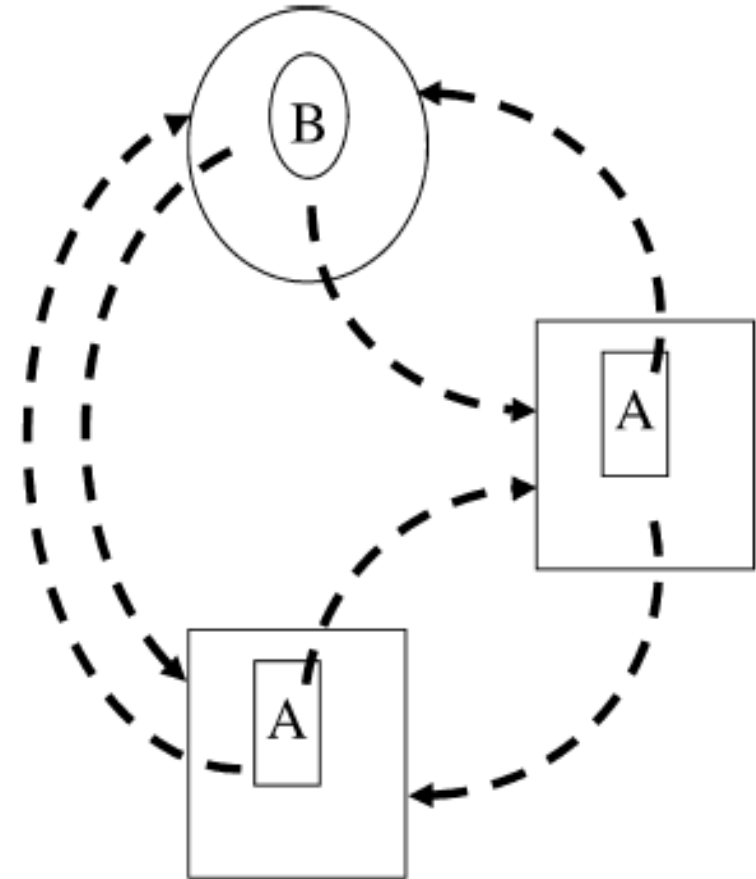
4 Paradigms:

- Species sorting
- Mass effects
- Neutrality
- Patch dynamics

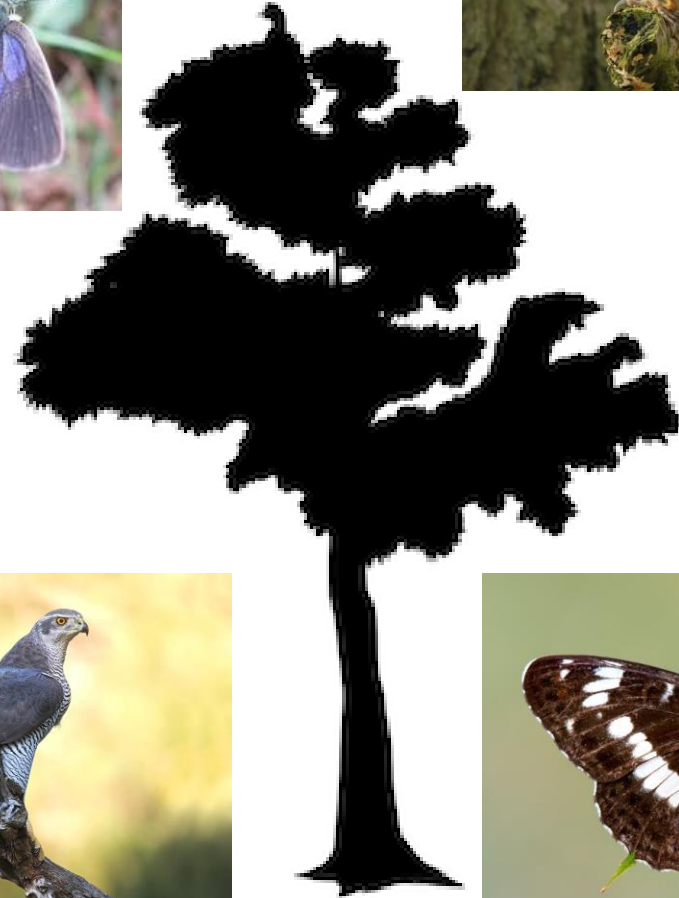
Species sorting

- Local **environmental conditions differ** among patches
- Species **differ in their niche**: their relative performance varies according to the environment
- There is **enough dispersal** so that the right species colonizes the right habitat

Species sorting results in a good match between local environmental conditions and species composition of local communities



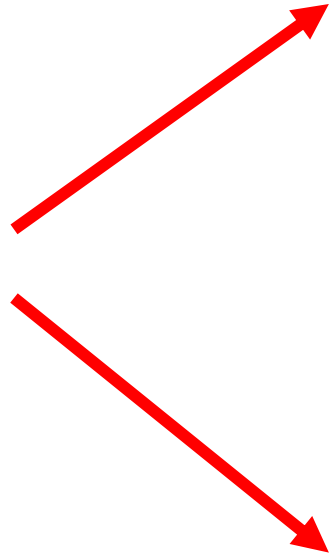
Species sorting



Your turn!



Butterflies are connected
to host plant!



[github.com/BramJanssens
1/SoS-Species-Sorting](https://github.com/BramJanssens/1/SoS-Species-Sorting)

Data collection

- 20 one hour observations (counting) of butterfly species
- Comparing butterfly communities of grassland vs. forest.
 - 10 grassland
 - 10 forest

Site_ID	Habitat	Species	Count
F1	Forest	Aphantopus hyperantus	4
F1	Forest	Pararge aegeria	4
F1	Forest	Gonepteryx rhamni	3
F1	Forest	Araschnia levana	12
F1	Forest	Polygonia c-album	2
F1	Forest	Aglais io	9
F1	Forest	Aglais urticae	7
F1	Forest	Vanessa atalanta	2
F1	Forest	Vanessa cardui	9
F2	Forest	Gonepteryx rhamni	4
F2	Forest	Pararge aegeria	8
F2	Forest	Polygonia c-album	10
F2	Forest	Araschnia levana	5
F2	Forest	Celastrina argiolus	13
F2	Forest	Aglais urticae	3
F2	Forest	Vanessa atalanta	6
F2	Forest	Vanessa cardui	4
F3	Forest	Araschnia levana	6

Statistical analysis

Carry out a PCoA (Principle Coordinate analysis)

- Construct community matrix

	Aphantopus hyperantus	Pararge aegeria	Gonepteryx rhamni	Araschnia levana	Polygonia c-album
F1	4	4	3	12	2
F2	0	8	4	5	10
F3	0	14	6	6	10
F4	12	4	2	1	0
F5	4	0	15	11	5
F6	1	0	13	14	13
F7	7	0	11	15	15
F8	2	14	0	13	1
F9	13	9	11	15	0
F10	14	11	14	11	5
G1	0	0	0	0	0
G2	0	0	0	0	0

Statistical analysis

Carry out a PCoA (Principle Coordinate analysis)

- Construct community matrix
- Calculate Bray-curtis distance
- Carry out PCoA with these distances

*Two points close together
are similar !*

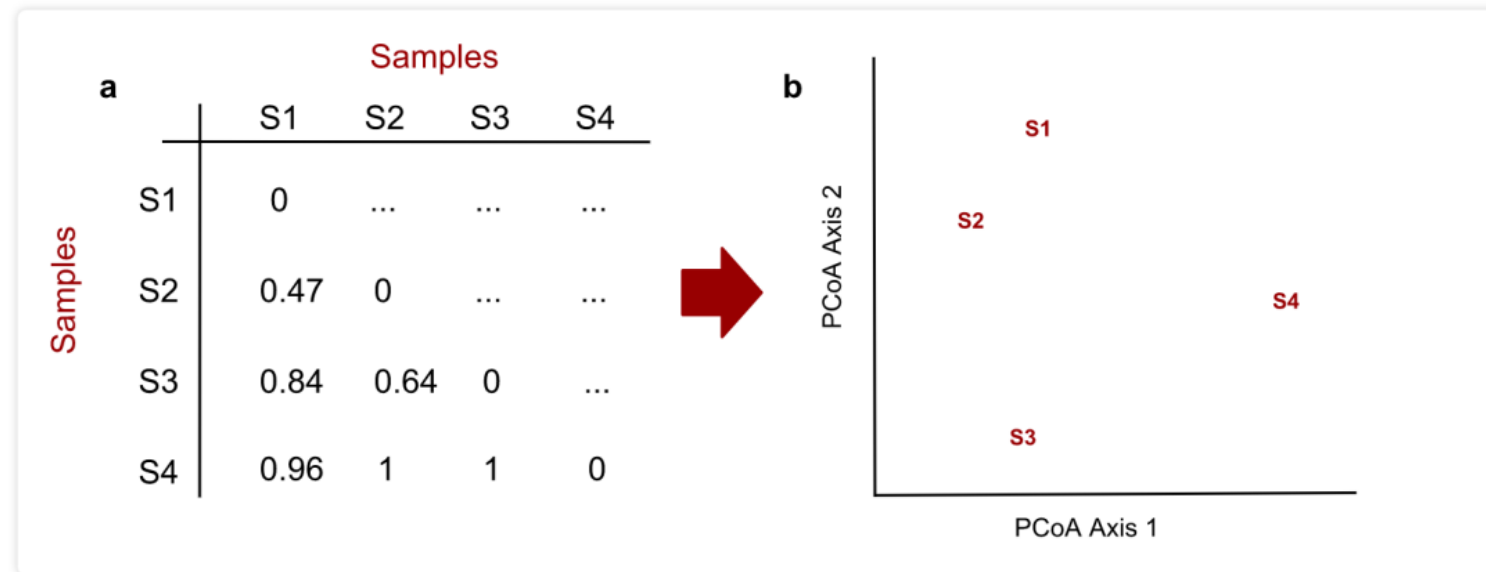
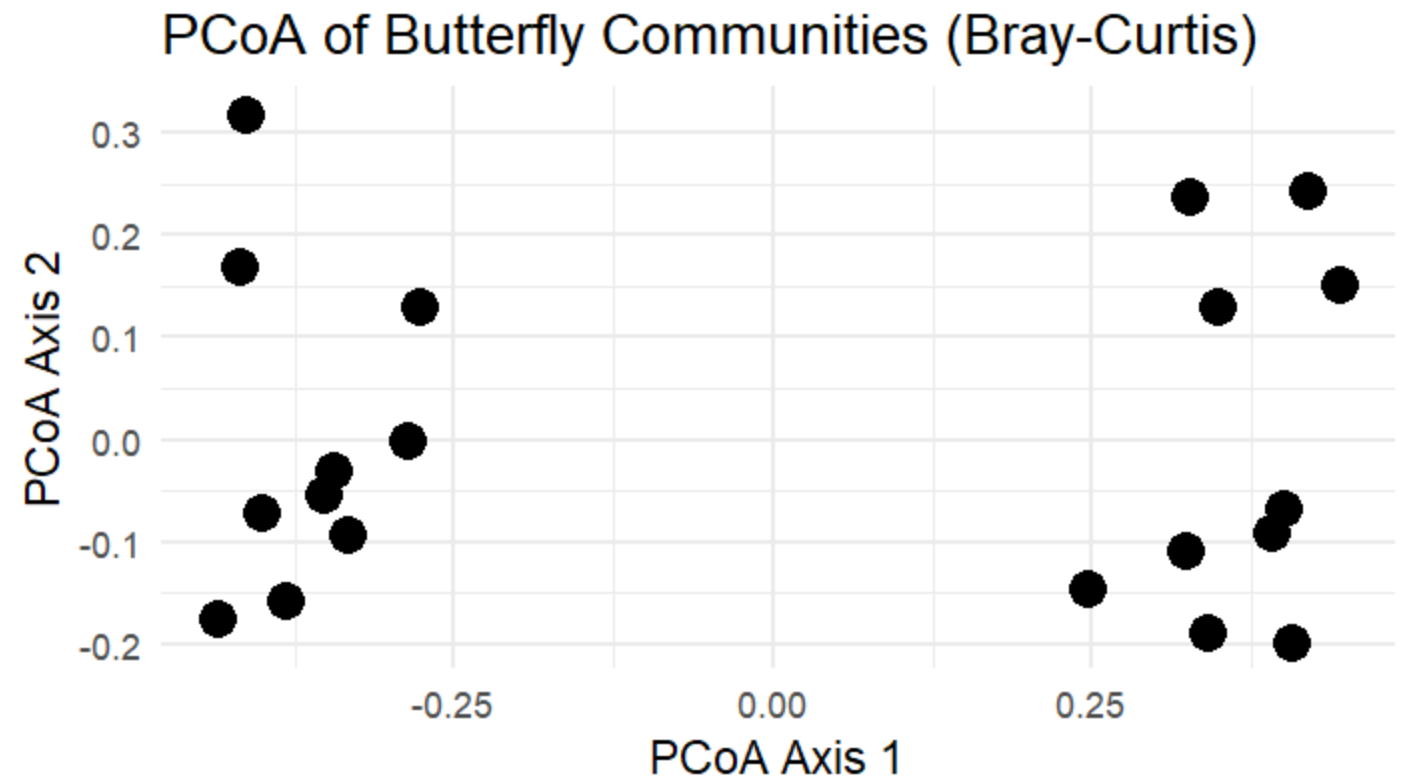


Figure 1: Principal coordinate analysis ordination of a Bray-Curtis dissimilarity matrix. Objects that are ordinated closer together have smaller dissimilarity values than those ordinated further apart. A successful PCoA will capture most of the variation in the (dis)similarity matrix in a few PCoA axes.

Statistical analysis

Carry out a PCoA (Principle Coordinate analysis)

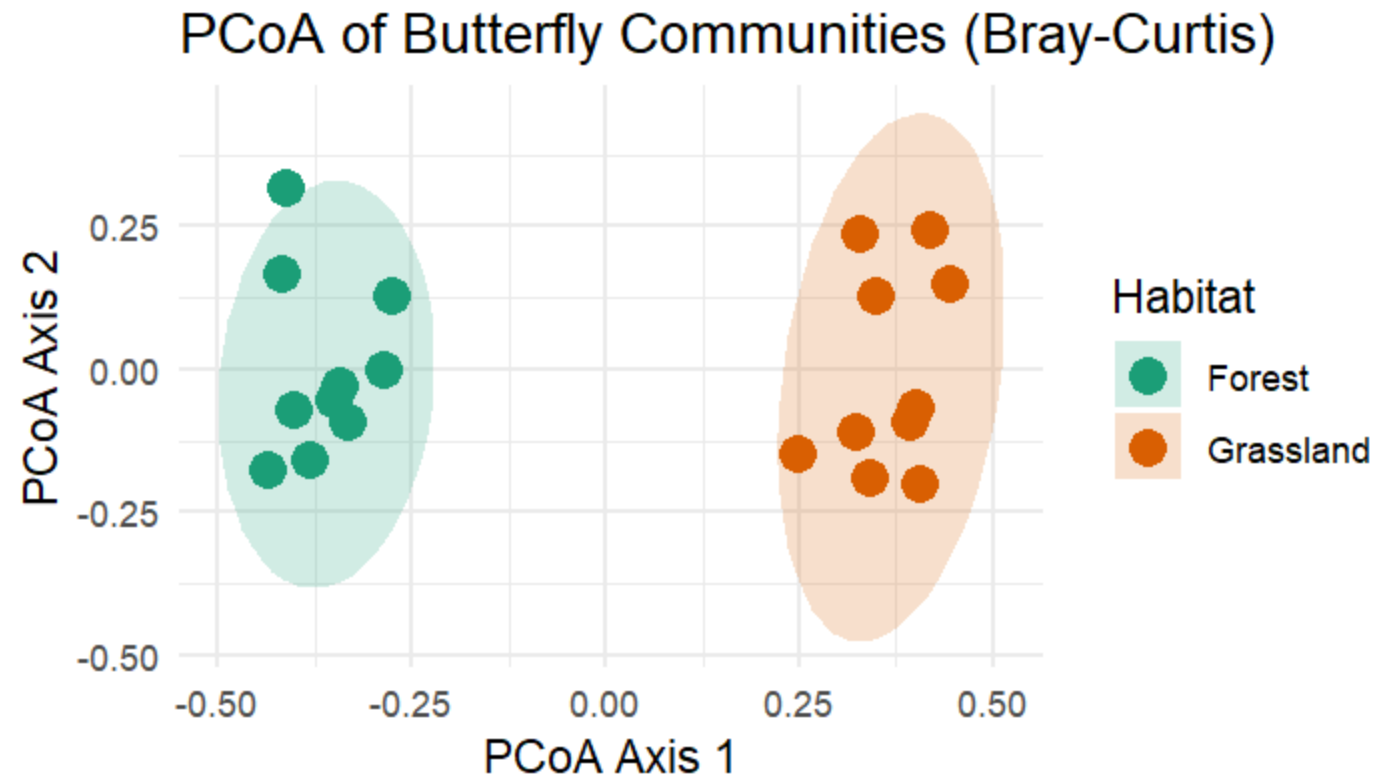
- Construct community matrix
- Calculate Bray-curtis distance
- Carry out PCoA with these distances
- Use data from PCoA to plot results



Statistical analysis

Carry out a PCoA (Principle Coordinate analysis)

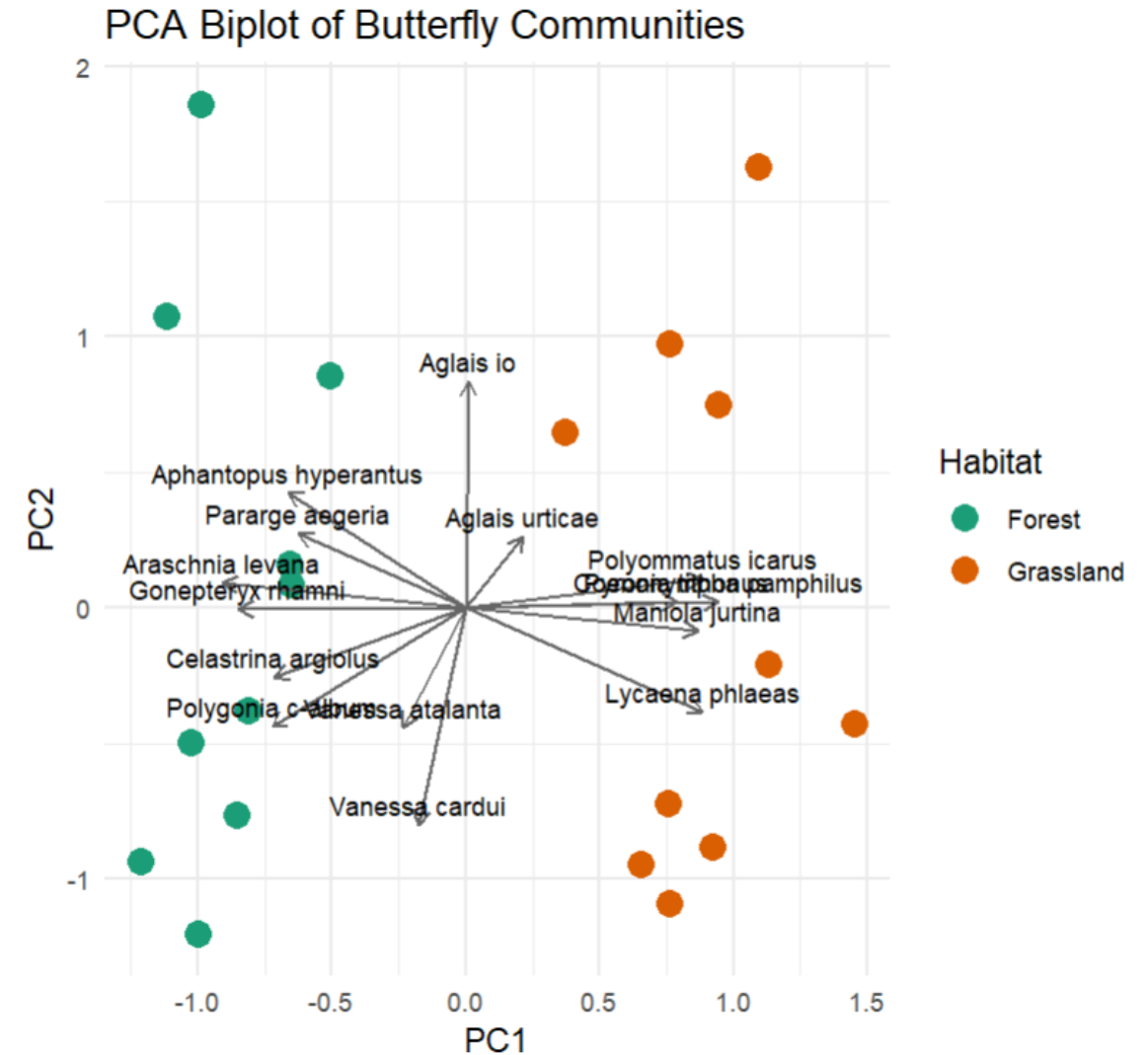
- Construct community matrix
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- Use data from PCoA to plot results



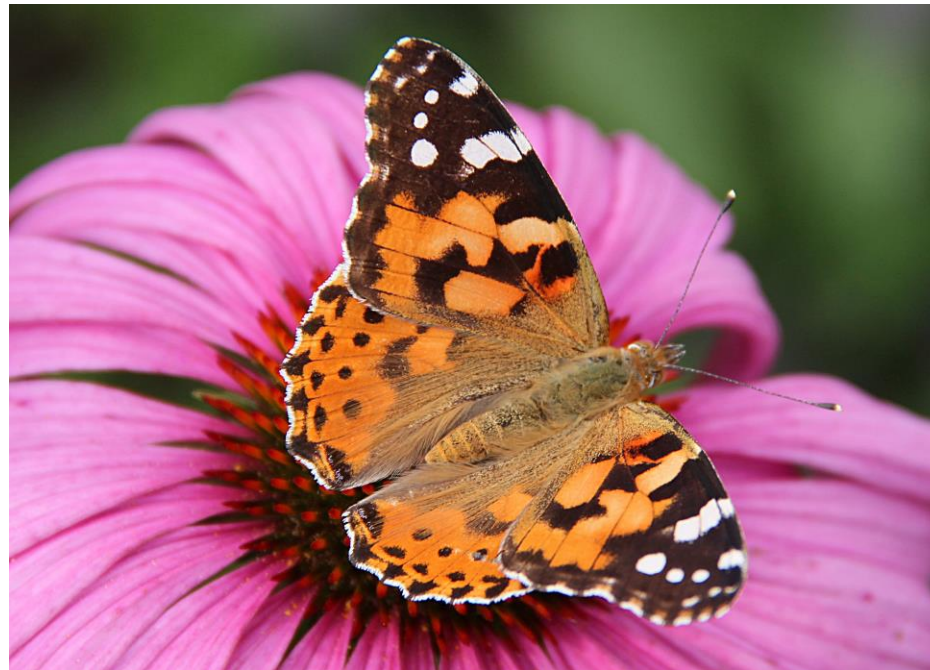
Statistical analysis

Carry out a PCoA (Principle Coordinate analysis)

- Construct community matrix
- Calculate Bray-curtis distance
- Carry out PCoA with these distances
- Use data from PCoA to plot results
- Which species are where?
 - PCA (able to add arrows)



Generalists vs specialists

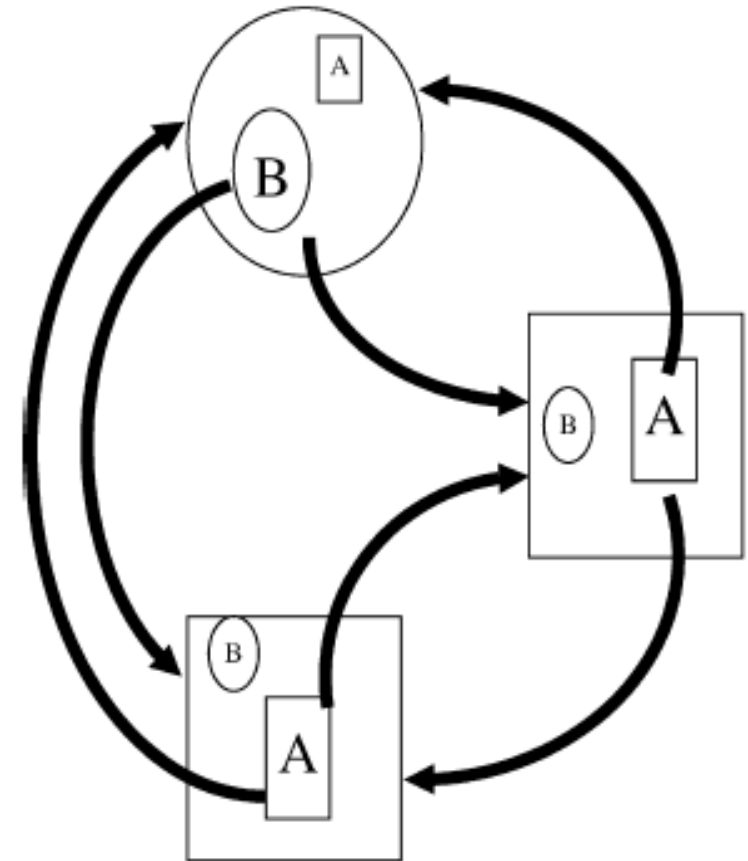


Mass effects

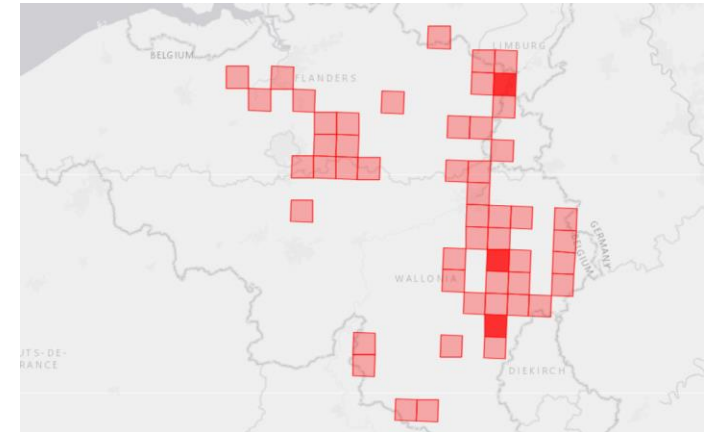
- **Patch quality differs**, and **high dispersal** rates result in source-sink dynamics
- High dispersal rates result in homogenization of the communities

Mass effects result in a weaker match between environment and species composition

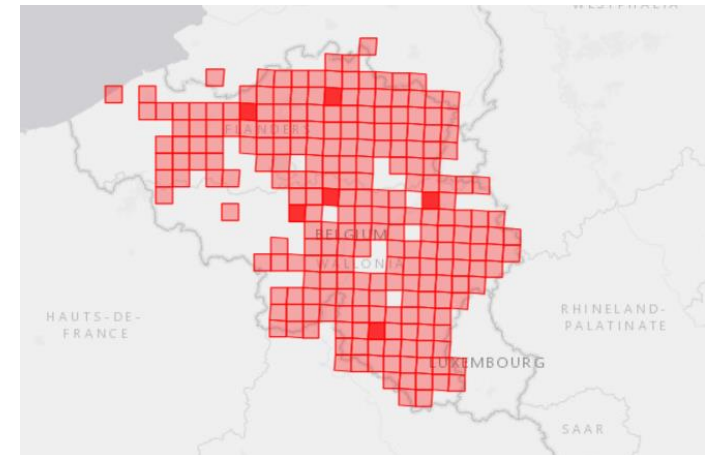
What effect does this have on overall diversity in a region?



Mass effects



2015



2025

Mass effects

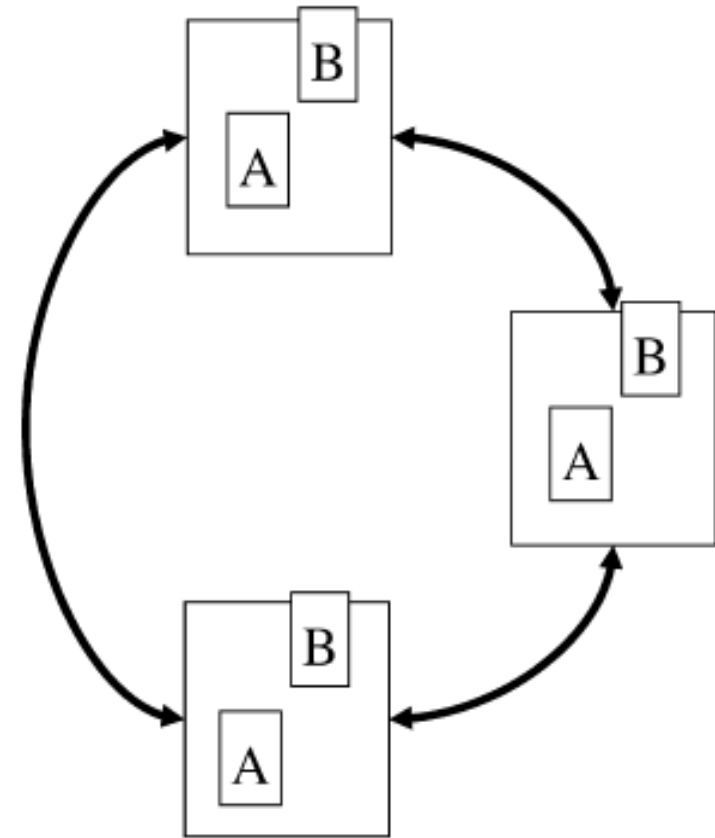
- Beaver in unfavorable habitat due to overcrowding



Neutrality

- All species are **ecologically equivalent**
- All **dynamics are fully neutral**, changes are due to ecological drift
- Stochastic extinctions are compensated by dispersal within the metacommunity
- Diversity at the metacommunity level is maintained by local speciation or from incoming dispersers from other metacommunities

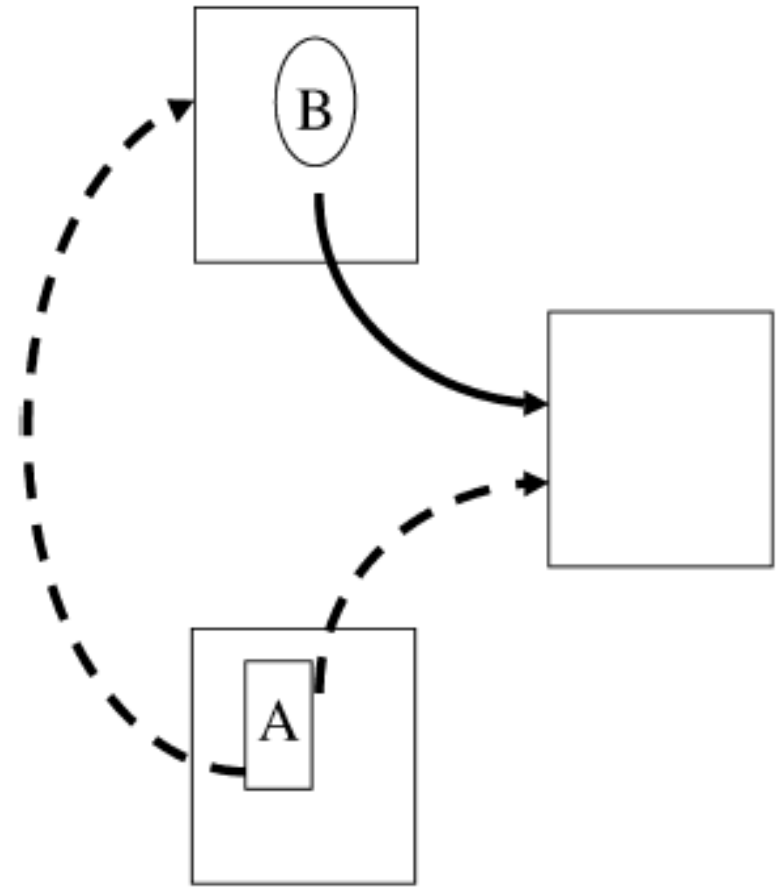
Predicts no match between the environment and species composition of local communities and differences due to ecological drift



Patch dynamics

- **Identical patches**
- Stochastic and deterministic extinctions
- Extinctions are countered by dispersal
- Species are characterized by tradeoffs between **dispersal ability** and **competitive strength**
- Intermediate levels of dispersal lead to regional coexistence

*Strong **temporal** dynamics*



Species B has high dispersal rate while species A is the stronger competitor in the patches

Patch dynamics



Patch dynamics



Evolving Metacommunities

What are evolving metacommunities?

- Traditional metacommunity models primarily consider:
 - species dispersal
 - environmental filtering
 - interspecific interactions

In natural systems, **evolutionary changes** can occur on **similar timescales** to **ecological processes**, resulting in influencing each other.

What are evolving metacommunities?

- Species traits can evolve in response to local environmental conditions and influence the interactions with other species.



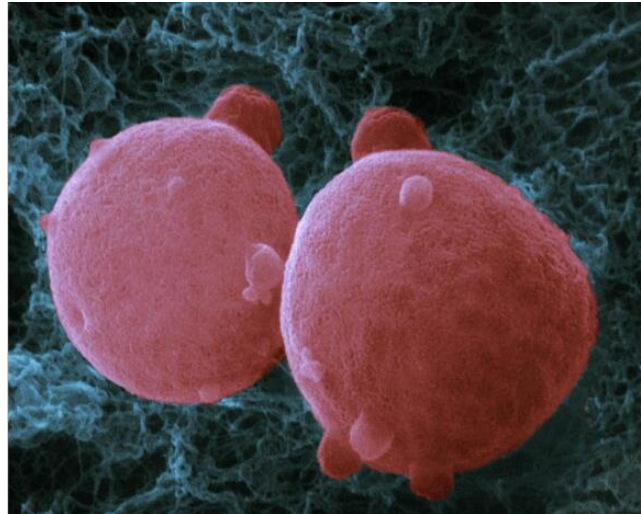
Evolving metacommunities: evolutionary dynamics affecting metacommunity dynamics and vice versa

Experimental study

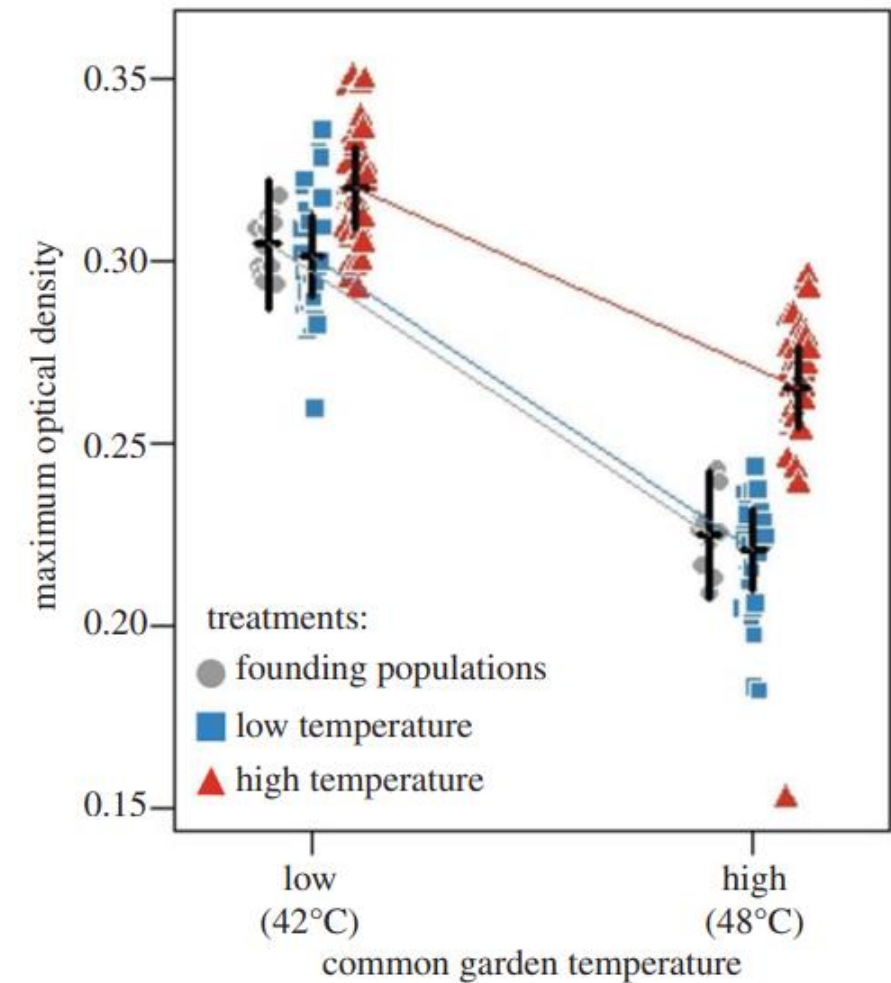
H. volcanii 135 days at high temperature (48°C)

- Higher densities when exposed to high temperature

Rapid evolution



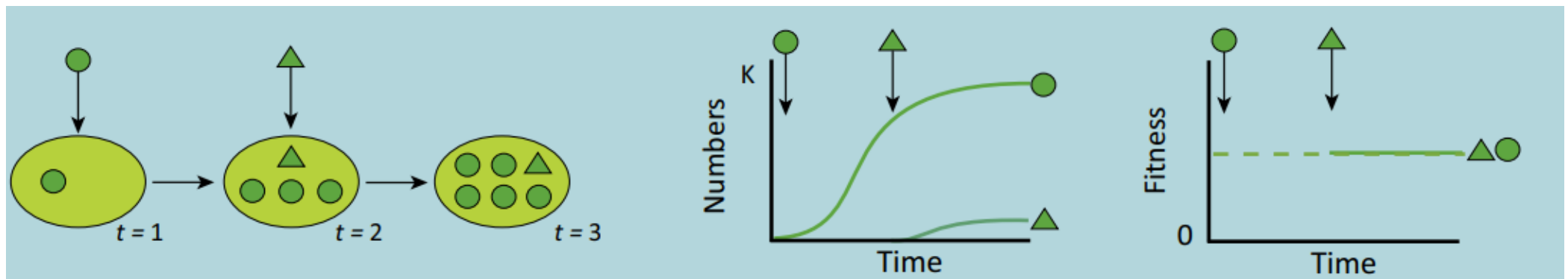
Remember this for later!



Nadeau et al., 2021

Priority effects

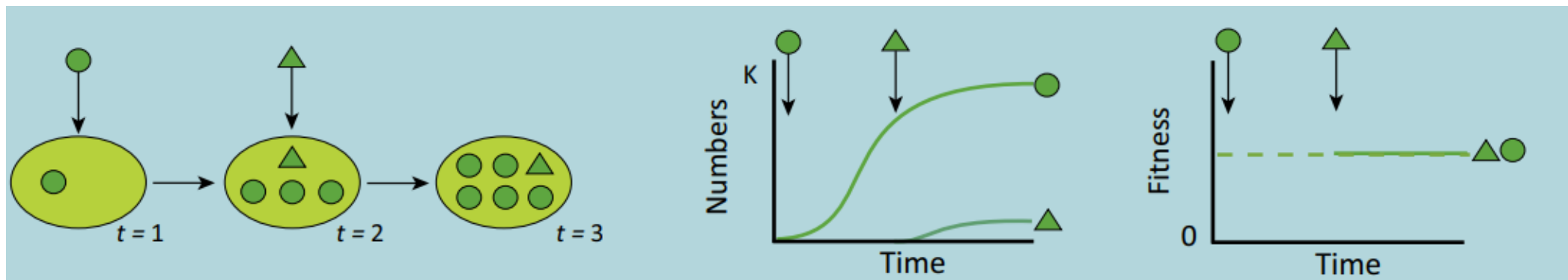
- **Priority effect:** the **order** in which species arrive at a location or are introduced into an environment can influence the composition and structure of the resulting community.



De Meester et al., 2016

Understanding figures

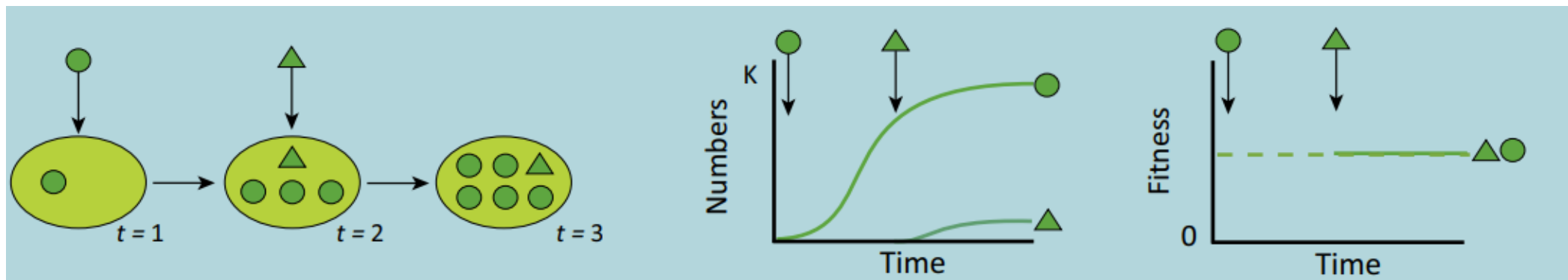
- **t** = timepoint
- **K** = **carrying capacity** (the maximum number of individuals a habitat can support sustainably)
- **Fitness** = a measure of how well a species/individual performs in a given habitat
- **Species** = ● and ▲
- **Habitat** = ○



De Meester et al., 2016

Priority effects

- Species/individuals have **different arrival times**
- Species/individuals have **similar fitness**
- No evolution
- **Monopolization** of resources -> **priority effects**

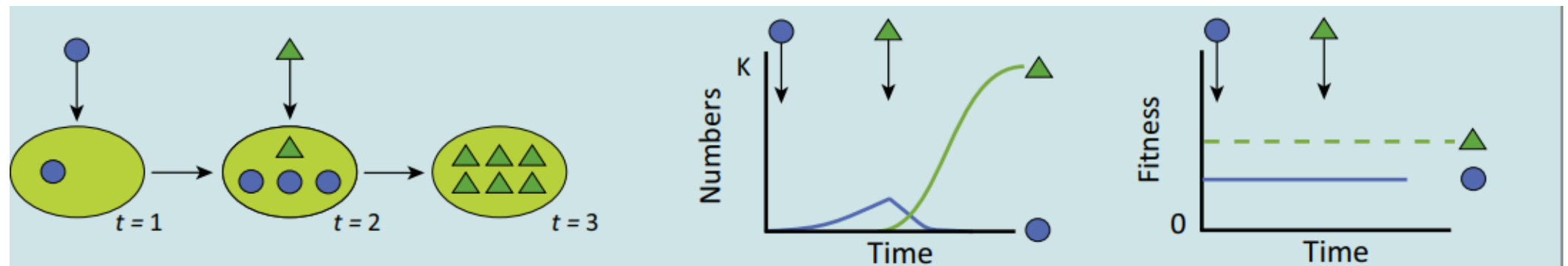


De Meester et al., 2016

Priority effects

- Species/individuals have **different arrival times**
- Species/individuals have **different fitness**
- No evolution
- First arriving species is outcompeted-> no **priority effects** and species sorting

Examples?



De Meester et al., 2016

Examples



Examples



Daphnia obtusa



Competition-colonization trade-off

- an ecological concept where species exhibit a **balance** between their ability to **compete** for resources within a habitat and their ability to **colonize** new habitats.



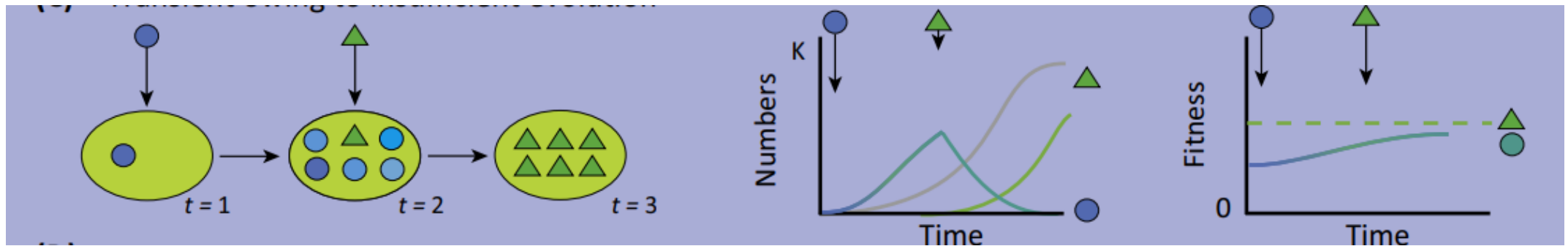
Priority effects

But! Evolution can be rapid!



Priority effects

- Species/individuals have **different arrival times**
- Species/individuals have **different fitness**
- **Evolution**, adaptation of first arriving species/individuals
- But **too slow** to not get outcompeted by subsequent species/individuals



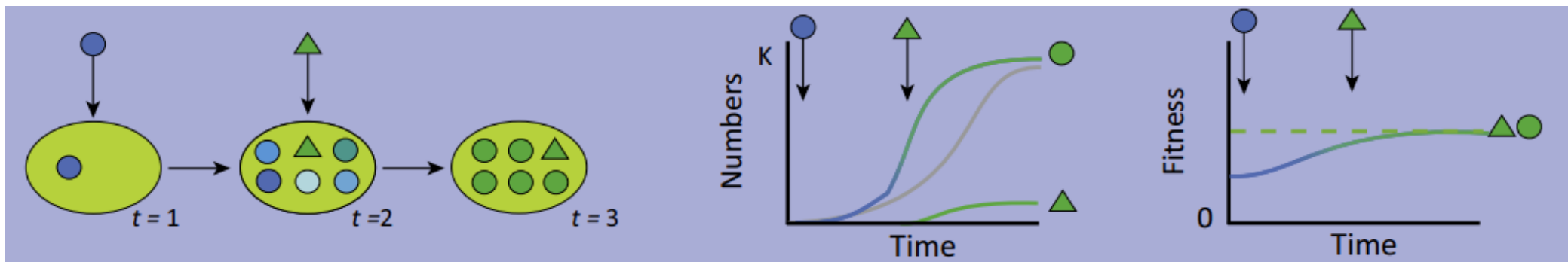
De Meester et al., 2016

Examples



Priority effects

- Species/individuals have **different arrival times**
- Species/individuals have **different fitness**
- **Evolution**, adaptation of first arriving species/individuals
- But **too slow** to **prevent colonization** by subsequent species/individuals but **coexistence**



De Meester et al., 2016

Example

- Experimentally altering arrival time of different waterflea species
- 60 days
- Competition experiment
- **Priority effects?**



Simocephalus vetulus



Daphnia obtusa

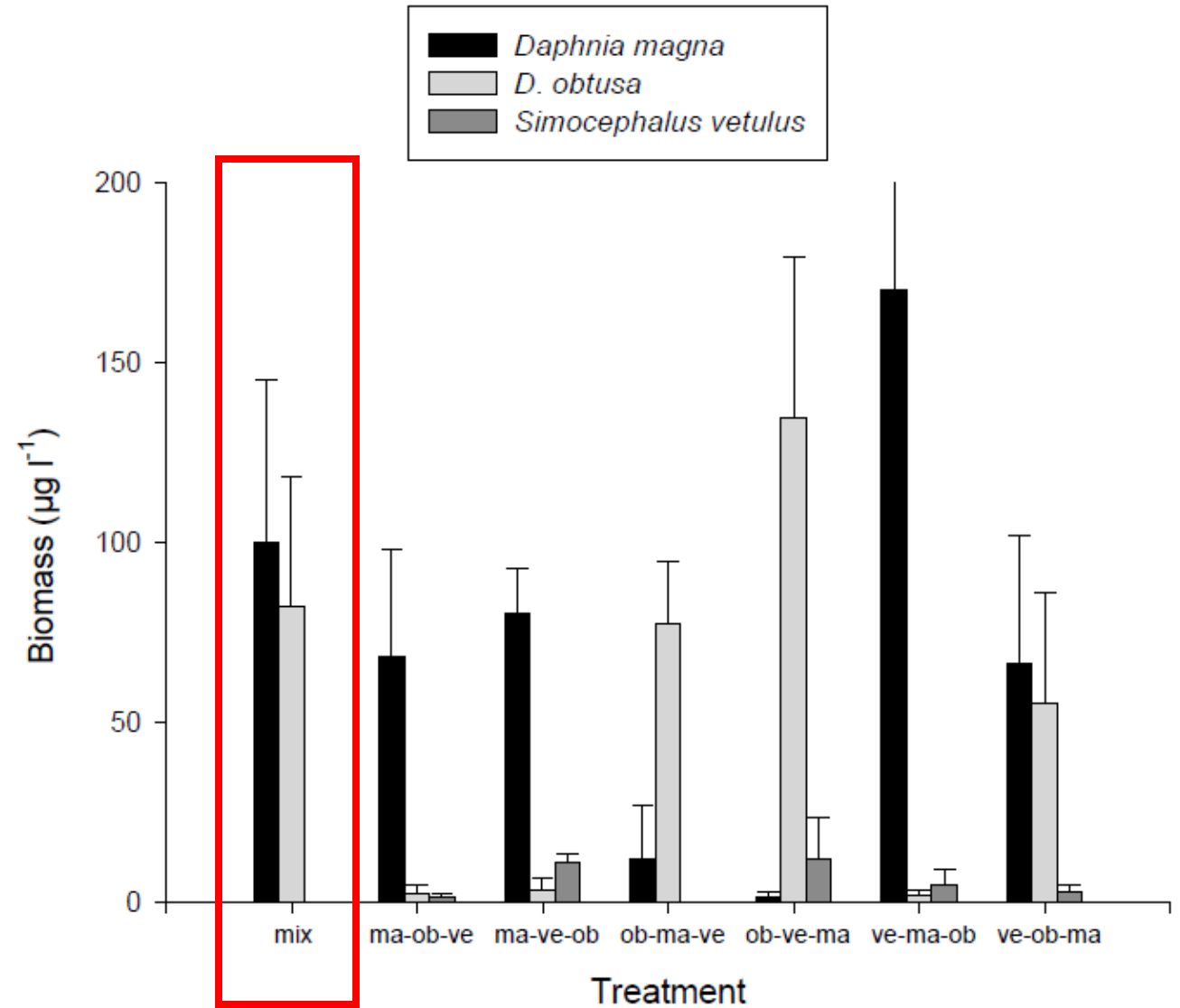


Daphnia magna

Louette & De Meester 2007

Mix

- *Daphnia magna* and *Daphnia obtusa* dominate
- *Simocephalus vetulus* is outcompeted

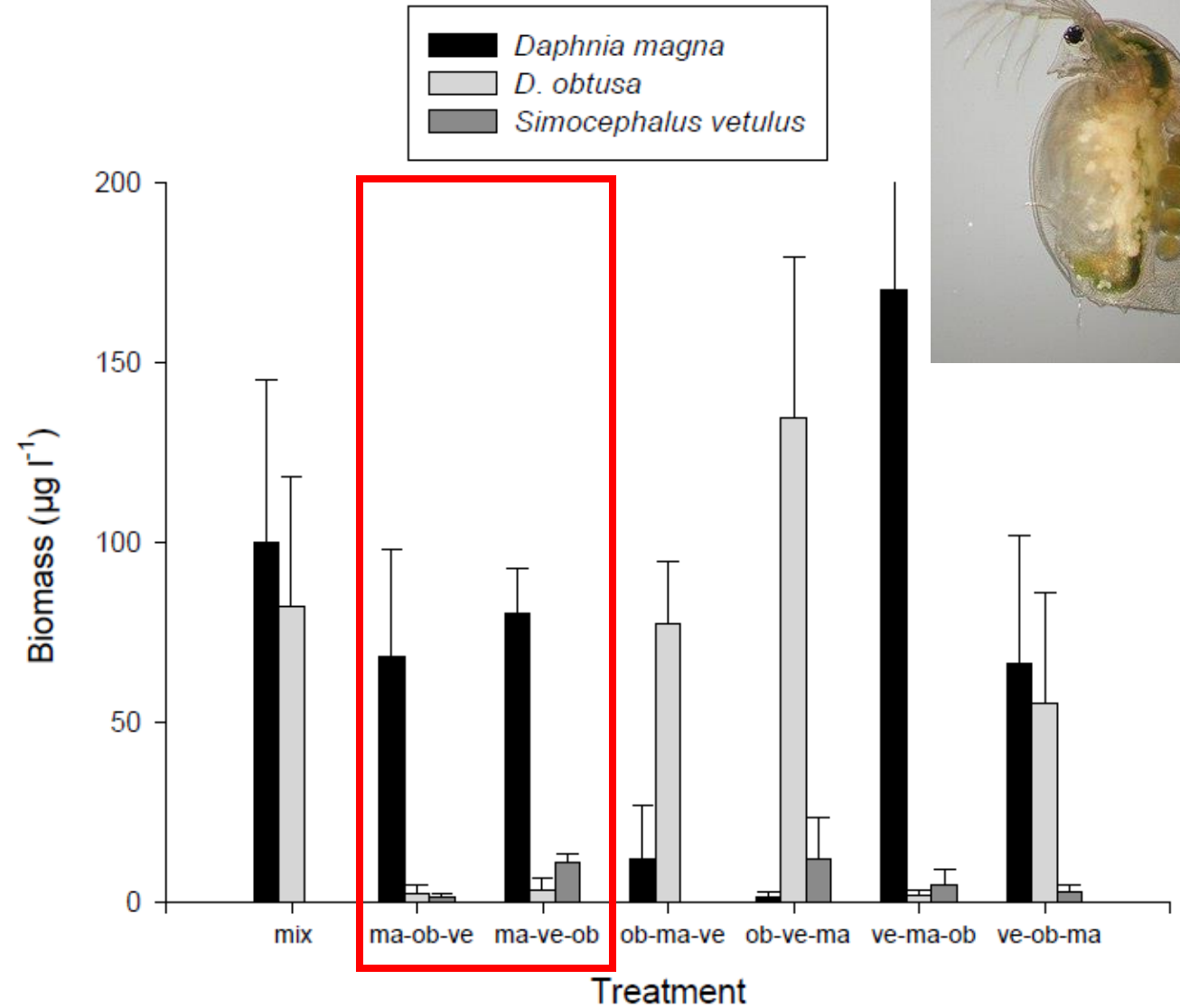
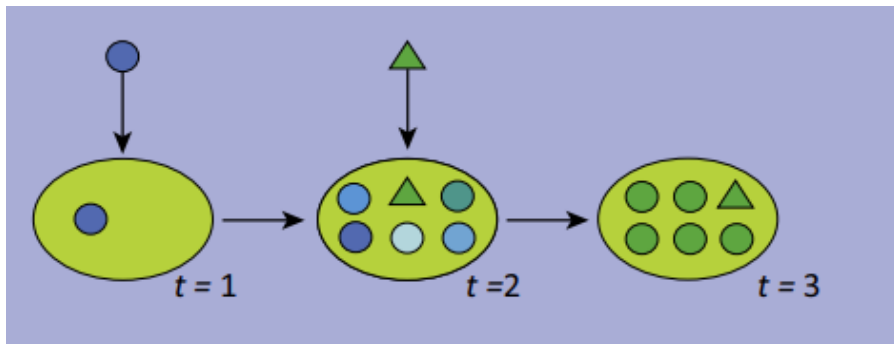


Louette & De Meester 2007

D. magna first

- *D. magna* dominates over *D. obtusa*

➤ Priority effects!

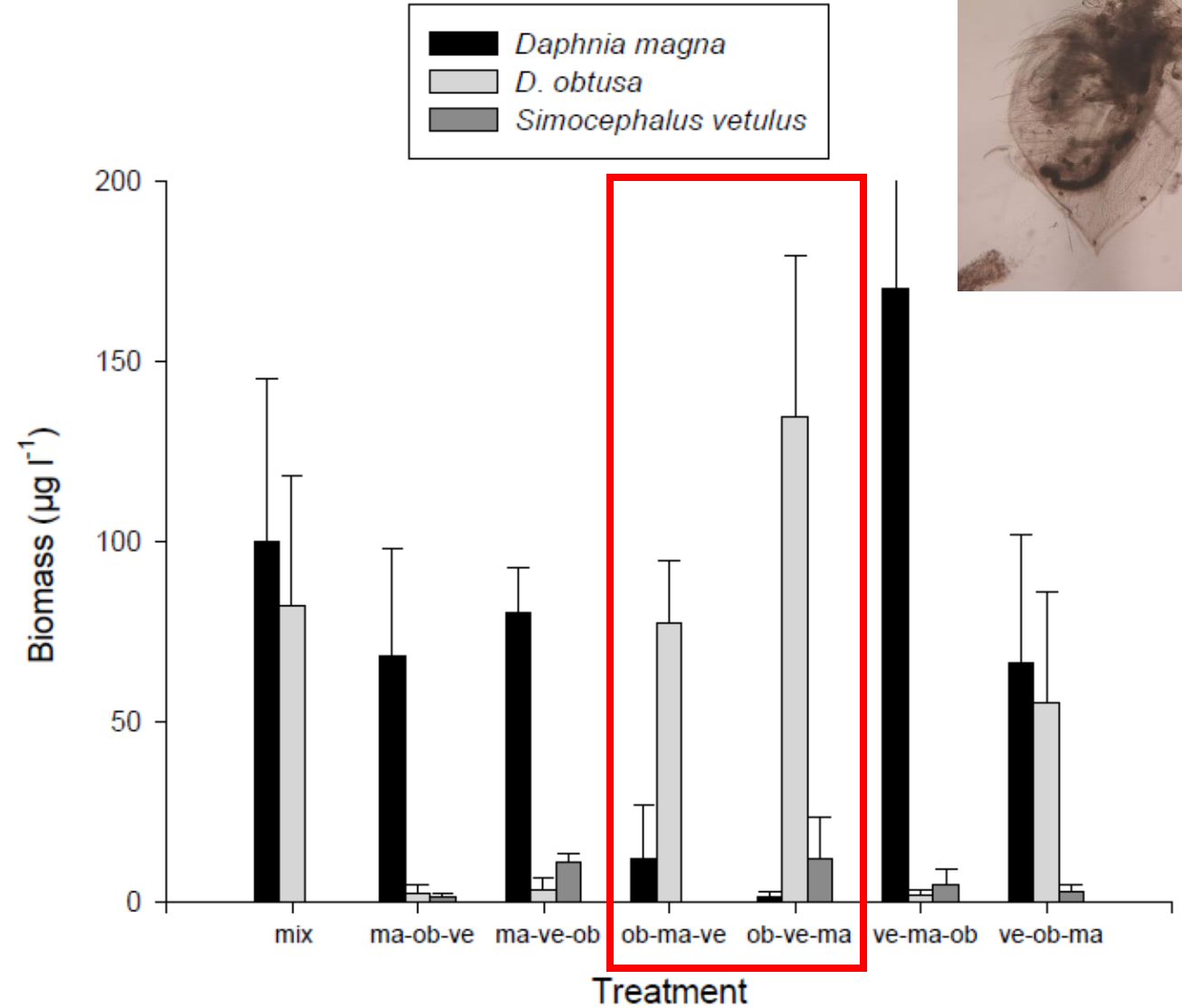
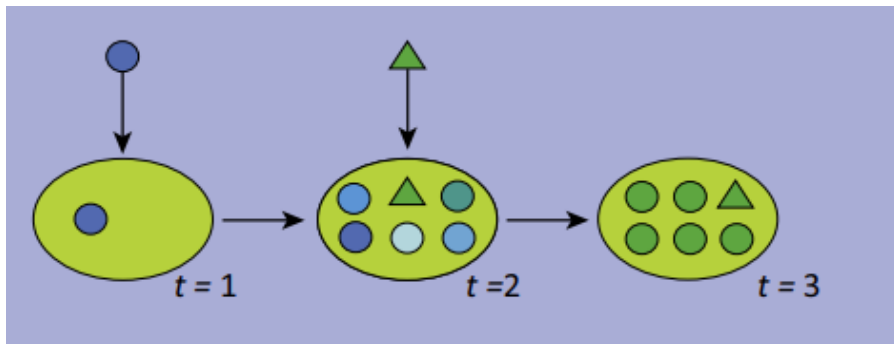


Louette & De Meester 2007

D. obtusa first

- *D. obtusa* dominates over *D. magna*

➤ Priority effects!

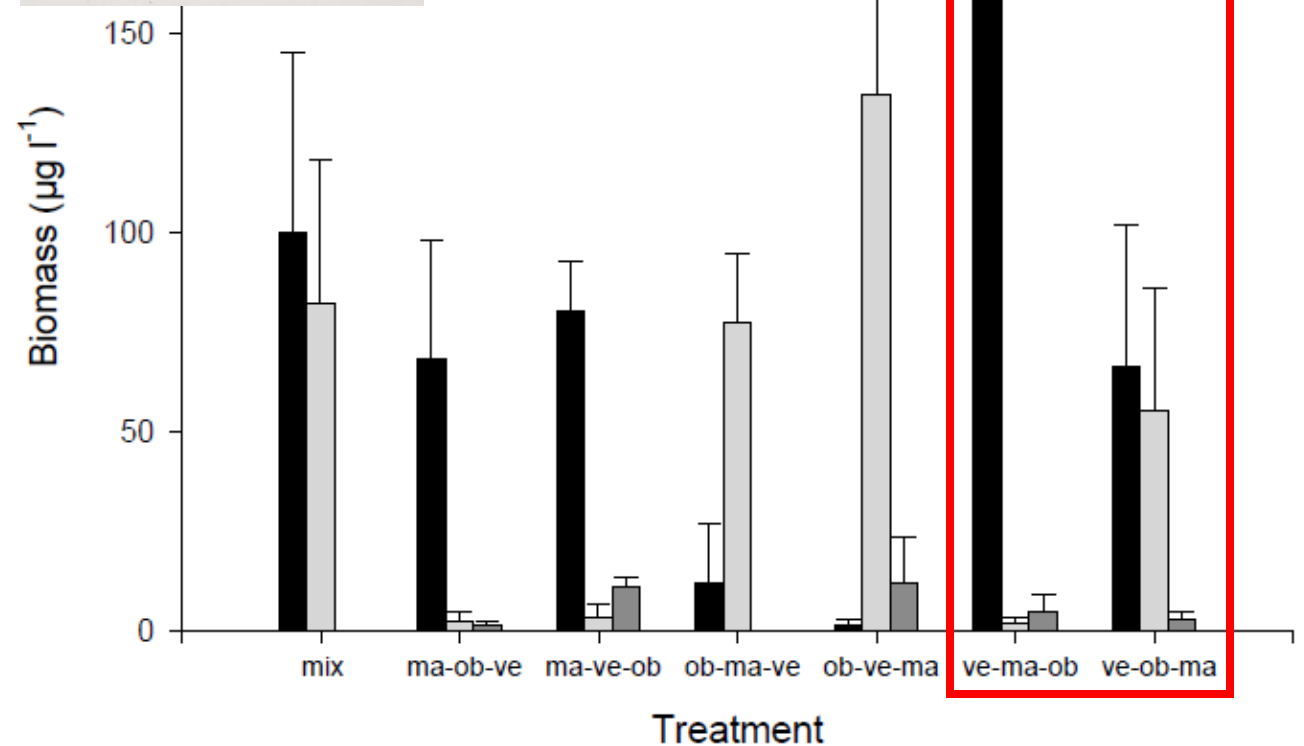
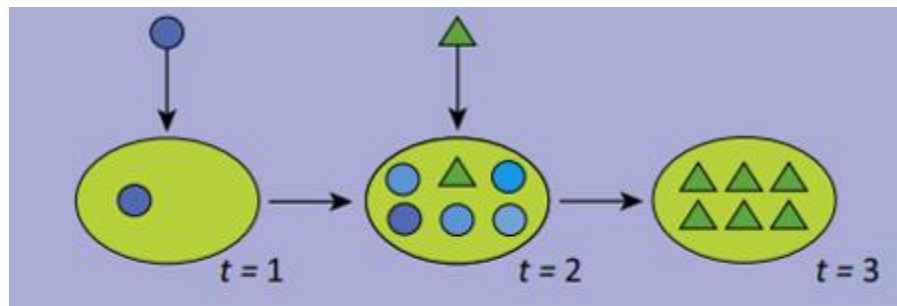


Louette & De Meester 2007

S. vetulus first

- *D. magna* dominates when secondly inoculated
- *D. magna* and *D. obtusa* co-exists when *D. obtusa* is secondly inoculated

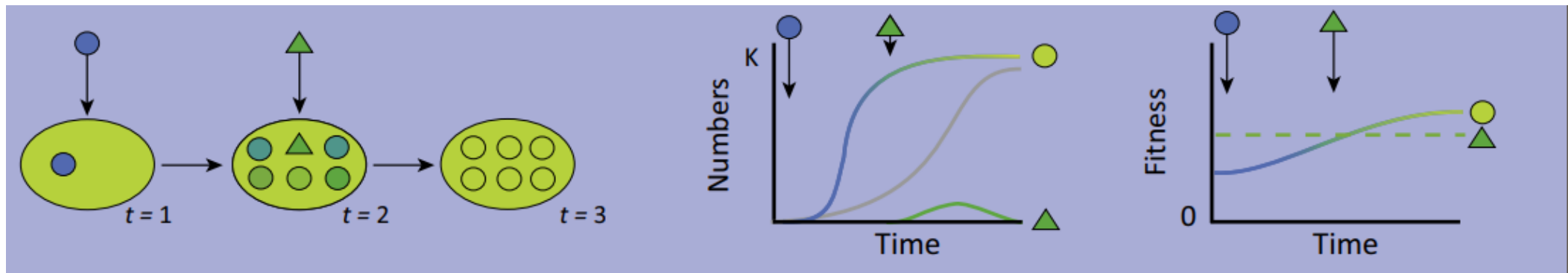
➤ No priority effects!



Louette & De Meester 2007

Priority effects

- Species/individuals have **different arrival times**
- Species/individuals have **different fitness**
- **Evolution**, adaptation of first arriving species/individuals
- **Rapid**, prevents colonization by subsequent species/individuals



De Meester et al., 2016

Evolution-mediated priority effects

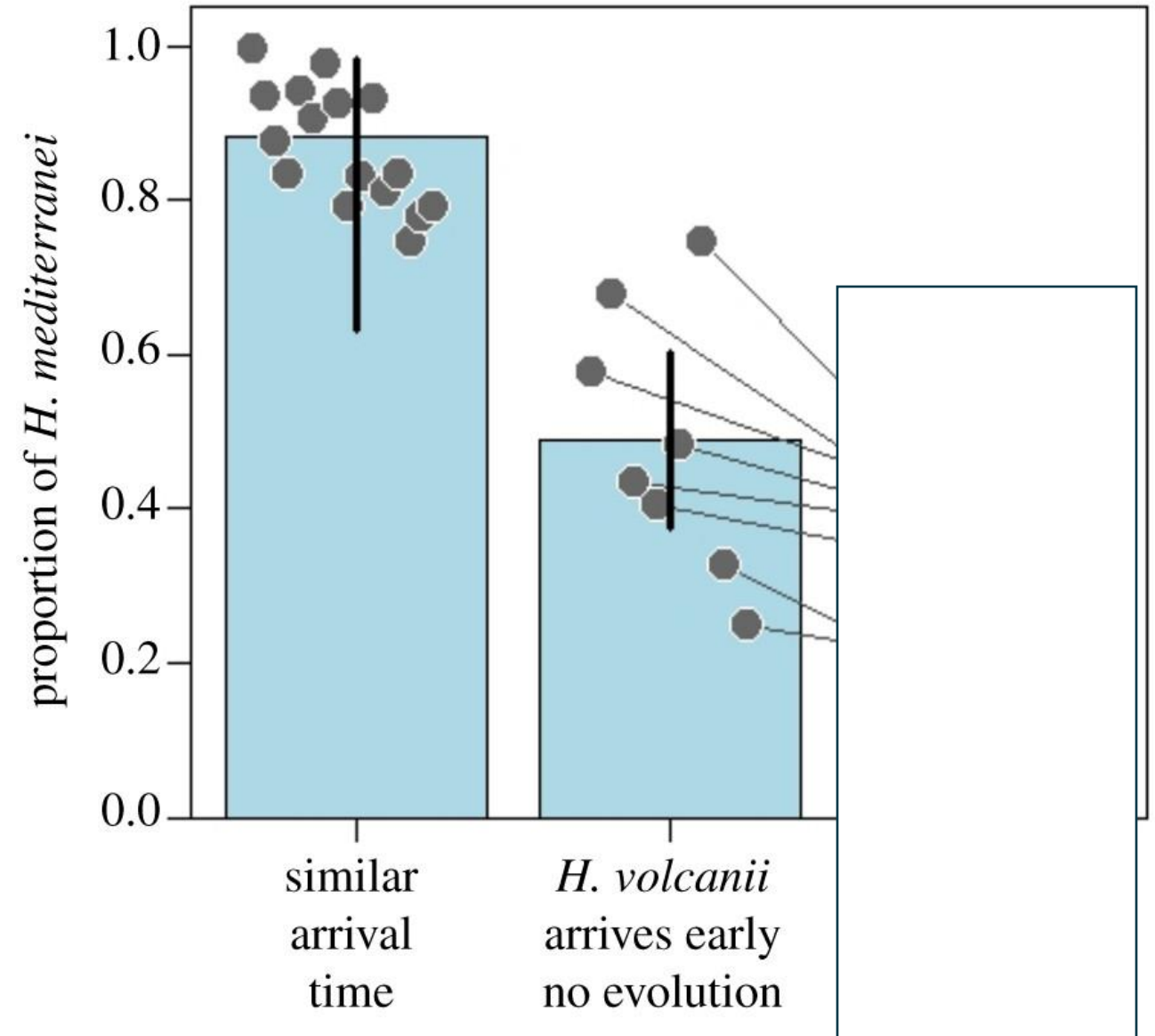
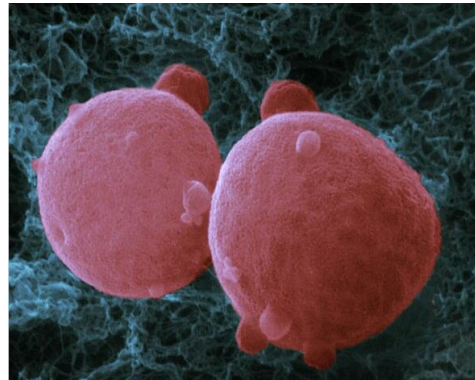
= The order in which species or genetic lineages arrive in a habitat, and how they evolve, influences subsequent community or population dynamics.

➤ **Rarely shown** in experiments due to multiple confounding factors

Example

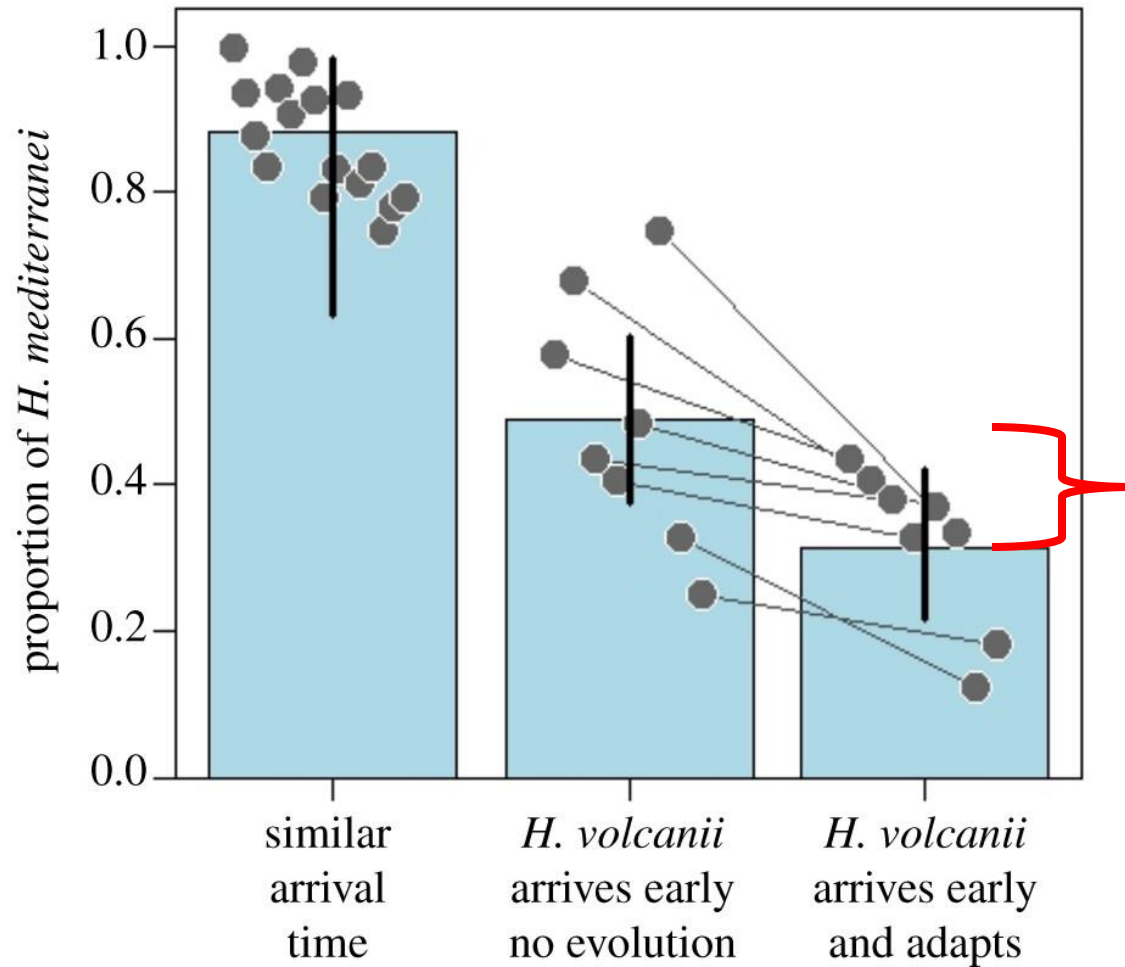
- Competition experiment of *H. volcanii* vs. *H. mediterranei*

➤ Priority effects!



Nadeau et al., 2021

Example



- If *H. volcanii* can evolve *H. mediterranei* will decrease even more in abundance

evolution

➤ **Evolution-mediated priority effects!**

Nadeau et al., 2021

Larger scales

